Sub Code: 3021 'B'

NEB GRADE XII

2079 (2022)

Grade Increment (Supplementary) Examination

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Time : 3 hrs. **Full Marks : 75**

Attempt all the questions.

Group 'A' **11 × 1 = 11**

Rewrite the correct options of each questions in your answer sheet.

1. In reaction of chloroform with primary amine in ethanolic solution of KOH is termed as
   a) Hoffmann's reaction b) Carbylamine reaction
   c) Reimer-Tiemann's reaction d) Kolbe's reaction
2. An organic compound (P) undergoes reduction with $LiAlH_4$ to produce (Q). When (Q) is heated with copper at 300°C, (P) is again produced. The compound (P) is
   a) $CH_3 - CHO$ b) $CH_3 - CO - CH_3$
   c) $CH_3CH_2 - OH$ d) $CH_3O.CH_3$
3. Which one is the laboratory test of alcohol?
   a) Iodoform test b) Victor-Mayer's test
   c) Esterification test d) Oxidation test
4. Chloroform does not give white ppt with silver nitrate because
   a) $AgNO_3$ is ionic compound.
   b) $CHCl_3$ does not ionize with water.
   c) $CHCl_3$ is chemically inert.
   d) $AgNO_3$ does not ionize with water
5. Vinegar contains
   a) 10-20% acetic acid b) 10% acetic acid
   c) 7-8% acetic acid d) 100% acetic acid
6. If a filter paper soaked by Hg^{++} ions is exposed in a gas jar containing NH_3 the filter paper will turn

a) Black b) Red c) Brown d) Blue

7. Which of the following compound used to silvering a mirror?

a) AgCl b) $AgNO_3$ c) AgS d) AgBr

8. Value of ionic product of water at 393 K is

a) Less than 1×10^{-14} b) Greater than 1×10^{-14}

c) Equal to 1×10^{-15} d) Equal to 1×10^{-7}

9. Which of the following conditions are correct?

a) $\Delta H = O, \Delta S < O$ b) $\Delta H > O, \Delta S > O$

c) $\Delta H < O, \Delta S < O$ d) $\Delta H > O, \Delta S < O$

10. If the weight of metal and chlorine in a metal chloride is 1:2, the equivalent weight of the metal is

a) 71 b) 35.5 c) 17.5 d) 8

11. For rate law experiment, two gases (X) and any Y are filled in a container. The rate law for the reaction has been found to be rate = $K[X]^2[Y]$. What will be the rate of reaction when pressure is doubled?

a) The rate will be doubled

b) The rate will becomes four times

c) The rate will becomes six times

d) The rate will becomes eight times

Group 'B' **8 × 5 = 40**

12. i) Give the statement of first law of thermodynamics.

ii) Deduce mathematical equation for it.

iii) Show that heat change at constant volume is change in internal energy.

iv) Mention a major limitation of first law of thermodynamics. [1+2+1+1]

Or

Standard hydrogen electrode is an electrode that scientists use for reference on all half-cell potential reactions.

i) What is meant by reference electrode?

ii) For the measurement of standard electrode potential of silver $E°Ag^+/Ag$, Hydrogen electrode is connected with it,

a) Represent an electrochemical cell notation.
b) Indicate cathode and anode.
c) If standard emf of the cell [E_o -cell] is 0.80v, calculate the standard electrode potential $E°Ag^+/Ag$. [2+1+1]

13. 0.70 gm of a sample of $Na_2CO_3.xH_2O$ were dissolved in water and volume was made to 100 ml. 20 ml of this solution required 19.8 ml of $\frac{N}{10}$ HCl for complete neutralization.
   i) Name the chemical indicator used in this titration.
   ii) Calculate numerical value of x.

14. The oxidation of different classes of alcohols given various products.
   i) Name mild and strong agents used in the oxidation of alcohol. [1]
   ii) Write down the oxidation products of 1°, 2° and 3° alcohols. [4]

15. For the following hypothetical reaction,
$$(A) \xrightarrow{Reaction\ I} (B) \xrightarrow{Reaction\ II} (C) \longrightarrow (D)$$
The compound (A) is an unsaturated aliphatic hydrocarbon and the compound (B) can be obtained by reduction of chlorobenzene with hydrogen in presence of Ni-Al/alkali on oxidation of (C) with CeO_2/H^+ give (D) as aromatic aldehyde.
   i) Use suitable reagent and condition in the reaction I and reaction II.
   ii) Identify (A), (B) and (C) giving proper chemical reaction. [2+3]

16. Write down an example of each of 1° amine, 2° amine and 3° amine giving IUPAC name. Give a test to distinguish 1° amine from 2° amine. [3+2]

17. You are given an organic compound having molecular formula C_3H_7X.
   i) Write down a primary and a secondary haloalkane giving their IUPAC name.
   ii) Give proper reaction sequence to convert the primary haloalkane into the secondary haloalkane.
   iii) What product is obtained, when the secondary haloalkane is subjected to Wurtz reaction?

iv) Give reason, why an haloalkanes more reactive than haloarenes towards nucleophic substitution reaction. [1+2+1+1]

Or

a) Write down correct reaction of each for the preparation of ethanoic acid from following sets of organic compounds.

i) CH_3CN ii) CH_3ON_3 iii) CH_3MgI

b) Convert ethanoic acid into methanoic acid. [1+1+1+2]

18. Write down the chemistry of blue vitriol. [5]

19. Zinc is considered as non-typical element and it belongs to the element of group II B

i) Why are the elements of group II B called volatile metal?

ii) Name the process of concentration of Zinc blende during the extraction of Zn.

iii) Write chemical reaction involved in roasting during extraction of zinc.

iv) What is meant by spelter zinc?

v) How is granulated zinc prepared? [5×1]

Group 'C' **3 × 8 = 24**

20. a) Write down correct example of each of the following reaction.

i) Wurtz reaction ii) Oxo process

iii) Esterification iv) Williamson's etherification

v) Diazotization vi) Tollens test

vii) Hell-Volhard Zelinsky reaction viii) Carbylamine reaction

Or

a) The given table shows the different organic compounds.

$A = C_6H_5NH_2$ $E = C_6H_5CH_3$

$B = C_6H_5OH$ $F = C_6H_5N_2Cl$

$C = C_6H_5CHO$

$D = C_6H_6$

Make a correct reaction sequence of the above compounds and proper conditions and reagent. [6]

b) Why is $-NH_2$ group protected before nitration in aniline? Write reaction for it. [2]

21. a) Cement is important materials for the constructions of building, road, bridge etc. [2+1+1+1+1]

i) Give a difference between hydraulic and non-hydraulic cement.

ii) List the compositions of Portland cement.

iii) What is the purpose of adding gypsum during cement production?

iv) Name any two instruments to check the quality of cement.

v) Which is the first established state owned cement industry in Nepal?

b) What is meant by artificial radioactivity? Write an example of it. [2]

22. a) Give proper reason.

i) NH_3 acts as Bronsted base but not Arrhenius base.

ii) NaCl can't undergo hydrolysis.

iii) Group II metal ions get precipitated as their sulphides when H_2S is passed through their acidified salts.

iv) Ostwald's dilution law is not applicable for strong electrolyte.

b) How many moles of $Ca(OH)_2$ must be dissolved to produce 250 ml of an queons solution of P^H 10.65?

Or

a) For a hypothetical reaction ($2A + 3B \rightarrow M + N$), Concentration of (M) increased by 0.02 ML^{-1} in 10 seconds. [1+1+1+1]

i) Write down rate expression for the above reaction.

ii) Calculate the rate of formation of (M).

Sub Code: 3021 'A'

NEB GRADE XII

2079 (2022)

Grade Increment (Supplementary) Examination

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Time : 3 hrs. **Full Marks : 75**

Attempt all the questions.

Group 'A' **11 × 1 = 11**

Rewrite the correct options of each questions in your answer sheet.

1. What is the normality of 0.3 M phosphorous acid (H_3PO_3)?
   a) 0.1 b) 0.3 c) 0.6 d) 0.9
2. The soluility product of CaF_2 is 3.4×10^{-11}. What is its solubility in 0.01M solution of NaF?
   a) 3.4×10^{-7} *mol l*$^{-1}$ b) 3.4×10^{-5} *mol l*$^{-1}$
   c) 3.4×10^{-2} *mol l*$^{-1}$ d) 3.4 *mol l*$^{-1}$
3. What would be the value of rate constant (k) if the concentration of reactant is increase by 'x'?
   a) $\ln \frac{k}{x}$ b) $\frac{k}{x}$ c) $k + x$ d) k
4. For the equilibrium reaction, $PCl_5 (g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
   Which of the following conditions are correct?
   a) ΔH = O, ΔS < O b) ΔH < O, ΔS < O
   c) ΔH > O, ΔS > O d) ΔH > O, ΔS < O
5. Functional isomer of methoxy ethane is,
   a) propanol b) propanal
   c) propanone d) propane
6. Which of the following metal is leached by cyanide process?
   a) Ag b) Na c) Al d) Cu
7. Nitrobenzene on reduction with Sn/HCl gives:
   a) 1, 3-dinitrobenzene b) nitrobenzene
   c) o-dinitrobenzene d) Aniline
8. What is the nature of the solvent used during the preparation of Grignard reagent?

a) alkyl hallide b) ester c) dry ether d) heat

9. Alcohol reacts with sodium metal then it gives sodium alkoxide and hydrogen gas it indicates:

a) Acidic nature of alcohol b) Basic nature of alcohol

c) Neutral nature of alcohol d) Amphoteric nature of alcohol

10. Ethanamide is obtained from ethanoic acid by the reaction of

a) Conc. H_2SO_4 b) P_2O_5

c) NH_3 d) Anhy. $AlCl_3$

11. The iron pipes carrying drinking water are covered with Zinc to prevent from rusting this process is called

a) allow formation b) Electroplating

c) Galvanization d) Electrifying

Group 'B' **8 × 5 = 40**

12. Concentration of sulphuric acid solution can be determined with titration with primary standard sodium carbonate solution.

a) Na_2CO_3 is considered as primary standard why?

b) Which chemical indicator is suitable for this titration?

c) If 10 cc of H_2SO_4 is completely neutralized by 17 cc of decinormal Na_2CO_3, what is normality of H_2SO_4? [1+1+3]

Or

Calculate the enthalpy change for the reaction

$H_2C = CH_2\ (g) + H_2(g) \rightarrow H_3C - CH_3(g)$ the bond energies of C–H, C–C, C = C and H–H are 99, 83, 147 and 104 K Cal/mol respectively. [5]

13. Define Zero-order reaction. The following data are given for the reaction

$2x + y \rightarrow$ Products.

Exp. No.	[X] $mol\ l^{-1}$	[Y] $mol\ l^{-1}$	Initial rate $mol\ l^{-1}s^{-1}$
1	0.1	0.1	7×10^{-3}
2	0.3	0.2	8.4×0^{-2}
3	0.3	0.4	3.36×10^{-1}
4	0.4	0.1	2.8×10^{-2}

Calculate:

i) The order with respect to X & Y.

ii) Half life of reaction with respect to X.

iii) Rate formation of product when $[X] = 0.6$ mol l^{-1} and $Y = 0.3$ mol l^{-1}.

14. Write any three important features of transition metal, Cu^{+} ion is transition metal but can't give colour why? [3+2]

15. Copper II sulphate crystals are commonly known as blue vitriol.

i) How would you obtain for saturated $CuSO_4$ solution? [1]

ii) Convert blue vitriol into red oxide. [2]

iii) Write any two applications of blue vitriol. [2]

16. Write down the Victor-Mayer's test of 1°, 2° and 3° alcohol. [5]

Or

A carbonyl compound (M) is used as nail polish remover. The compound (M) contains three carbon atoms and it undergoes iodoform test.

i) Identify the compound (M).

ii) Write down a chemical reaction for the preparation of M.

iii) How is (M) converted into propane?

iv) Predict the final product obtained when (M) is heated with CH_3MgI in presence of dry ether and following by hydrolysis.

v) Give a laboratory test reaction of carbonyl compound. [5×1]

17. $C_6H_5OH \xrightarrow{A} C_6H_6 \xrightarrow{B} C_6H_5CH_3 \xrightarrow{C} C_6H_5CHO$

Choose current reagent (A), (B) and (C) in the above reaction sequence. How would you convert C_5H_6 into $C_6H_5N_2Cl_2$? [3+2]

18. Write down any three method of preparation ethanoic acid. How is ethanoic acid converted into ethanamine? [3+2]

19. Write an example of each of the following reaction.

i) Rosenmund's reduction

ii) Sandmayer's reaction

iii) Dehydrohologination

iv) Carbylamine reaction

v) Markovnikov's rule

Group 'C' [3 × 8 = 24]

20. a) Derive relationship between P^H and P^{OH} and calculate the P^H of 10^{-3} M KOH. [4]

b) Write the application of common ion effect and solubility product principle in qualitative salt analysis. [4]

Or

a) Calculate the enthalpy of combustion of methane if enthalpy of formation of methane, water and carbondioxide are –440, –72 and 93 KJ. [4]

b) Standard electrode potential for the following electrode reaction at standard state is given [4]

$Zn (s) \rightarrow Zn^{++}(aq) + 2e$, $E° Zn/2n^{++} = 0.76$ V

$Cu^{++}(a) + 2e \rightarrow Cu(s)$, $E° Cu^{++}/Cu = 0.34$ V

i) Calculate emf of cell.

ii) What would be the ΔG for the reaction?

iii) Can we store $CuSO_4$ solution in Zn-container?

iv) Write the cell notation for the reaction.

21. a) A sweet smelling organic compound (A) slowly oxidized by air in the presence of sunlight to give highly poisonous gas carbonyl chloride.

i) Why above compound (A) stored in dark and brown bottle?

ii) Give principle reaction involved in the preparation of compound (A) from ethanol.

iii) How would you convert compound (A) into (a) Chloretone (b) Chloropicrin (c) Ethyne.

iv) Why the compound (A) cannot give white ppt whit $AgNO_3$ solution?

[1+2+3+2]

Or

Give proper reason

i) phenol is more acidic than aliphatic alcohol.

ii) amino group is protected before nitration of aniline.

iii) Grignard reagent is stored in dry ether.

iv) Williamson's etherification is useful for the preparation of both symmetrical and unsymmetrial ether. [2+2+2+2]

22. a) What is meant by artificial radioactivity? Write an example of it. 2

b) PVC and nylon-6,6 are two common polymers widely used in daily life.

i) State the process of polymerization by which nylon-66 is formed.

ii) Write down the chemical reaction to form PVC.

c) i) Write down the importance of cement industry in Nepal.

ii) Define the term clinker.

Sub.Code : 3021'D'

NEB-GRADE XII

2079 (2022)

Chemistry

New course

For regular students (except technical stream)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Time: 3 hrs. **Full Marks: 75**

Attempt all the questions.

Group 'A' **11× 1=11**

Rewrite the correct options of each question in your answer sheet.

1. $2R - OH + 2Na \rightarrow 2RONa + H_2$, is an example of
   a) acidic nature of alcohol
   b) basic nature of alcohol
   c) electrophilic substitution reaction
   d) nucleophilic substitution reaction
2. $C_6H_5CH_2 - CHO$ and C_6H_5CHO can be distinguished by
   a) Iodoform test b) Tollen's test
   c) 2,4-DNP test d) Fehling test
3. An organic compound (X) undergoes reduction with $LiAlH_4$ to yield (Y). When vapour of y are passed over freshly reduced copper at 300°C (X) is formed. The compound Y is
   a) CH_3CHO b) CH_3CH_2-OH
   c) CH_3-CO-CH_3 d) CH_3-O-CH_3
4. The number of possible structural isomers of 1° amines of molecular formula $C_4H_{11}N$ give
   a) 1 b) 2 c) 3 d) 4
5. Acetic anhydride is obtained from acetyl chloride by the reaction of
   a) Conc. H_2SO_4 b) P_2O_5
   c) CH_3COONa d) CH_3COOH
6. A metal (M) forms thiosulphate with molecular formula $M_2S_2O_3$. The valency of metal is
   a) 1 b) 2 c) 3 d) 4
7. The P^{11} of 10^{-8} M aqueous solution of HCl is

a) less than 7 b) 7 c) 8 d) more than 8

8. A catalyst accelerates the reaction because
   a) it brings reactants closer
   b) it lowers the activation energy
   c) it increases the activation energy
   d) it forms complex with the reaction

9. What is the concentration of nitrate ions if equal volume of 1M $NaNO_3$ and 1M KCl are mixed?
   a) 0.25M b) 0.5M c) 1M d) 2M

10. Tailing of mercury is due to the formation of
    a) HgO b) Hg_2O c) HgO_2 d) Hg_2O_2

11. Bell metal is an alloy of
    a) Cu, Pb and Sn b) Sn and Cu
    c) Zn and Pb d) Zn, Cu and Sn

Group 'B'

8×5=40

12. An electrochemical cell is constructed by using aluminum and silver electrodes whose electrodes potential values are
    $E°\ Al^{3+}/Al = -1.67V$
    $E°\ Ag^{+}/Ag = 0.8V$
    i) What is meant by electrochemical cell?
    ii) Represent an electrochemical cell using above electrodes.
    iii) Write down complete cell reactions.
    iv) Calculate the emf of the cell. (1+1+2+1)

Or

Hess's law is applied to calculate the different types of enthalpy of reactions.
i) Define Hess's law of constant heat summation.
ii) What is meant by enthalpy of reaction?
iii) Standard enthalpy of formation of $H_2O_2(l)$ and are $H_2O\ (l)$ are-188 KJ/mol and 286 KJ/mol respectively. What will be the enthalpy change of the following reaction.
$2H_2O_2(l) \rightarrow 2H_2O\ (l) + O_2(g)$

13. Ionic product of water at 25°C is 1×10^{-14} and water is regarded as very weak electrolyte.

i) Define ionic product of water.

ii) Deduce the relation $K_w = [H^+] [OH]$

iii) Calculate the $[OH^-]$ concentration of 0.01M HCl at 25°C.

iv) What is the effect of temperature on ionic product of water?

(1+1+2+1)

14. Ethyl alcohol is common alcohol and is used to manufacture alcoholic beverage. It can be prepared from sugar containing materials like molassess by fermentation process.

(1+1+1+1+1)

i) Define fermentation

ii) What is meant by molassess?

iii) Mention the function of yeast in the fermentation of ethyl alcohol.

iv) Write chemical reaction for the conversion of cane-sugar into ethyl alcohol.

v) Give a difference between absolute alcohol and **denaturated** alcohol.

15. A carbonyl compound (M) contains three carbon atoms and it undergoes iodoform test. (5 × 1)

i) Identify the compound (M).

ii) Write down a chemical reaction for the preparation of (M).

iii) How is (M) converted into propane?

iv) Predict the final product obtained when (M) is treated with CH_3MgI in presence of dry ether and followed by hydrolysis?

v) Give a laboratory test reaction of carbonyl compound.

Or

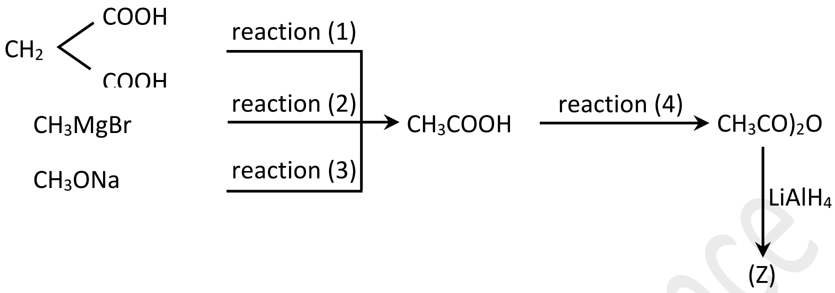
Convert ethoxy ethane from a halo alkane C_2H_5Br by using Williamson's reaction.

i) What product is obtained when ethoxyethane is heated with excess HI?

ii) Why are old sample of ethe not distilled to dryness in air?

iii) Convert phenol into anisole. (1+2+2)

16. For the following reaction sequence.



i) Write down **reagents** and conditions for reaction (1), reaction (2), reaction (3) and reaction (4).

ii) Identify the compound (Z) giving IUPAC name. (4 + 1)

17. How would you apply Hoffmann's method for the separation of 1°, 2° and 3° amine from their mixture?

18. An important compound non typical transition metal zinc which is used as eye lotion and is also called white vitriol. (5×1)

i) Write down a method of preparation of white vitriol.

ii) What happens when white vitriol is heated to 800°C?

iii) Define double salt giving an example of it.

iv) How is Lithopone obtained from white vitriol.

v) Why is zinc considered as non-typical transition metal.

19. Steel manufactured by Open Hearth process. (1+2+1+1)

i) What is Open Hearth process?

ii) Write down the chemical reactions occurring in Open-Hearth furnace.

iii) Why is Spiegeleisen added in the Open-Hearth furnace?

iv) Write down the composition of stainless steel.

Group 'C' **3×8 = 24**

20. a) Write an example of each of the following reactions.

i) Hydroboration oxidation

ii) Decarbonylation

iii) Sandmeyer's reaction

iv) Iodoform reaction
v) Elimination reaction
vi) Cannizzaro's reaction
vii) Reimer-Tiemann reaction
viii) Fridel Craft alkylation

Or

An unsaturated hydrocarbon (C_3H_6) undergoes Markovikov's rule to give (A). Compound (A) is hydrolysed with aqueous alkali to yield (B). When (B) is treated with PBr_3, compound (C) is produced. (C) reacts with AgCN (alc.) to give another compound (D). The compound (D) if reduced with $LiAlH_4$, produce (E).

i) Define Markovikov's rule.
ii) Identify (A), (B), (C), (D) and (E) with chemical reaction.
iii) How does E react with nitrous acid?
iv) How would you convert (B) into C_3H_8? (1+5+1+1)

21. a) For a hypothetical chemical reaction $mP + nQ \rightarrow z$; the rate Law is, rate = K $[P]^m$ $[Q]^n$. Where K is rate constant of the reaction (m + n) are overall order. (4×1)

i) Define rate law.
ii) Why is rate law experimental parameter?
iii) What is meant by rate constant?
iv) Mention a difference between order and molecularity of a reaction.

b) For the above reaction, the order of reaction with respect to P and Q are the first order and zero order respectively. Experimental data obtained from the reactions are: as below.

Expt.	[P]M	[Q]M	Initial rate (M sec^{-1})
I	0.1	0.1	2×10^{-2}
II	(A)	0.2	4×10^{-2}
III	0.4	0.4	(C)
IV	(B)	0.2	2×10^{-2}

i) Identify the value of A, B and C.
ii) Calculate rate constant [k]. (3+1)

Or

a) Crystal of oxalic acid is generally used to prepare primary standard solution. (1+1+1+1)

i) Define primary standard solution.

ii) Which chemical indicator is used in the titration of $KMnO_4$ solution verses oxalic acid solution?

iii) Why is oxalic acid solution warmed adding dilute H_2SO_4 before titrating with $KMnO_4$?

iv) Mention a major application of titration in quality control laboratory.

b) An aqueous solution of a dibasic acid containing 17.7 gm of acid per litre of the solution, has density 1.0077 gm/litre (molar mass of the acid = 118 gm/mol) Calculate; i) molarity ii) molality. (2+2)

22. a)

i) What is Portland cement?

ii) Name the major components present in Portland cement.

iii) Why is gypsum used in clinker during cement production process?

iv) Give any two instruments used for the quality control of cement.

(1+1+1+1)

b) i) Differentiate between homo-polymer and co-polymer

Sub.Code : 3021 'C'

NEB-GRADE XII

2079 (2022)

Chemistry

New course

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Time: 3 hrs. **Full Marks: 75**

Attempt all the questions.

Group 'A' **11× 1=11**

Rewrite the correct options of each questions in your answer sheet.

1. Alcohol vapour can be dehydrated by passing over.

   A) Conc. H_2SO_4 B) heated Al_2O_3

   C) heated Cu D) anhydrous $CaCl_2$

2. (A) $\xrightarrow{aq.NaOH}$ (B) $\xrightarrow{K_2Cr_2O_7/H^+}$ (C).

   The compound (C) undergoes Clemenson's reduction to give propane. The compound (A) is

   A) Propan-1-ol B) Propan-2-ol

   C) 2-halopropane D) Propanoic acid

3. Phenol on treatment with conc. HNO_3 in the presence of Conc.H_2SO_4 gives

   A) m-nitrophenol B) o-nitrophenol

   C) p-nitrophenol D) Picric acid

4. Acetaldehyde and acetophenone can be separated from their mixture by using the reagent.

   A) $NaOH+I_2$ B) 2, 4- DNPH

   C) $NaHSO_3$ D) NH_2-NH_2

5. The order of basic strength of 1° amine, 2° amine, 3° amine and ammonia is

   A) $3^o>2^o>1^o>NH_3$ B) $2^o>1^o>3^o>NH_3$

   C) $2^o>3^o>1^o>NH_3$ D) $3^o>1^o>2^o>NH_3$

6. Which element is added with copper to form bronze ?

   A) Fe B) Mn C) Sn D) Zn

7. During the extraction of iron from hematite the limestone acts as;
   A) Reducing agent B) Gangue
   C) Flux D) Slag
8. Which one is the strongest Bronsted base in the following ions ?
   A) ClO^- B) ClO_2^- C) ClO_3^- D) ClO_4^-
9. The unit of rate constant is (k) mol Lit^{-1} sec^{-1}, the reaction is.
   A) Zero order B) first order
   C) Pseudo first order$^-$ D) 2nd order
10. The heat of reaction in the following chemical reaction
    $2H_2(g)+O_2(g) \rightarrow 2H_2O(l)$ is termed as
    A) enthalpy of formation B) enthalpy reaction
    C) enthalpy of combustion D) enthalpy of solution

11. How much water should be evaporated from 400 ml of $\frac{N}{HCl}$ HCl to make it
    exactly 2N.
    A) 360 ml B) 370ml C) 380ml D) 390ml

Group 'B'

8×5=40

12. Acidic strength of solution can be measured in term of P^H value.
    i) Define pH of a solution
    ii) What is the application of pH in our daily life?
    iii) If equal volume of two solutions having (pH = 1) and (pH = 12) are
    mixed together, what will be the P^H of the solution? (1+1+3)

Or

State and explain the Hess's law of constant heat summation. (5)

13. Standard hydrogen electrode can be used as reference electrode. The standard electrode potential of standard hydrogen electrode is taken as zero.
    i) Define standard hydrogen electrode
    ii) What is the major application of standard hydrogen electrode?
    iii) Draw well labeled diagram of standard hydrogen electrode.

iv) What is meant by standard electrode potential?
(1+1+2+1)

14. Grignard reagent is an organometallic compound which is used to synthesizevarious organic compound. Starting from CH_3MgBr, how would you prepare

i) methane ii) ethanoic acid
iii) ethanol iv) propan-2-0l
v) 2-methyl propan-2-ol

Or

An organic compound (A) when heated with Ag powder gives C_2H_2 and form carbonyl chloride when it exposes to air.

i) Identify the compound (A)
ii) Write reaction for the laboratory method of preparation of (A)
iii) What happens when the compound (A) is treated with conc.nitric acid.
iv) convert (A) into methanoic acid (1+2+1+1)

15. Give principle and procedure for the separation of 1°, 2° and 3° amines by
Hoffmann's method.
(5)

16. A carbonyl compound (X) gives iodoform test and can be obtained by oxidation of a monohydric alcohol containing three carbon atoms.

i) Write down the structural formula of (X).
ii) Give functional isomer of (X) giving IUPAC name. iii) How is (X) prepared from 2,2- dichloro propane.
iv) Write reactions for the conversion of (X) into 2-hydroxy-2methyl propanoic acid.
v) What is the laboratory test of carbonyl compound? (5×1=5)

17. Complete the following reaction sequences using suitable reagents and conditions in P, Q, R, S and T

$$CH_3CH_2OH \xrightarrow{(P)} CH_3COOH$$

$$CH_3COOH \xrightarrow{Q} CH_3COOCH_3$$

$$CH_3COOH \xrightarrow{R} \quad \xrightarrow{S} CH_4$$

$$CH_3COOH \xrightarrow{T} ClCH_2COOH$$

(5×1=5)

18. Penta-hydrated copper sulphate is called blue vitriol.
   i) Starting from metallic, copper, how can you obtain blue vitriol ?
   ii) What happens when aq.solution of blue vitriol is treated with excess ammonia solution ?
   iii) Give chemical reaction of conversion of blue vitriol into black oxide.
   iv) Why is hydrated copper sulphate called blue vitriol ? (1+1+2+1)

19. A metal (M) is extracted from its suphide ore whose atomic number is 80. It also occurs as the amalgams of certain metals and is popularly known as quick silver.
   (5×1=5)
   i) Name the metal 'M' and write down the molecular formula of its common ore.
   ii) Why is this metal called quick silver.
   iii) State the process by which the ore is concentrated during its extraction.
   iv) Write down the chemical reaction involved during reduction in its extraction.
   v) Mention a major use of the metal.

Group 'C' **3×8=24**

20. a) Nitrobenzene is an industrial chemical and yellow oily liquid with almond like odour.
   i) Write down the chemical reaction for the nitration of benzene to prepare nitrobenzene.
   ii) Why is nitro group in nitrobenzene meta-directing in nature?
   iii) What happens when nitrobenzene is reduced with electrolytic medium.
   iv) Give a structural formula of an aromatic poly nitro compound which is used as explosives. (1+2+1+1)

   b) Consider the following reaction sequence.

$$C_6H_5CH_3 \xrightarrow{(x)} C_6H_5CHO \xrightarrow{alc.\ KCN} B$$

i) Write down reagent and condition in place of (x).

ii) Identify the compound (B)

iii) What product is obtained when C_6H_5CHO is subjected to Clemenson's reduction? (1+1+1)

Or

Given an example of each of the following reactions.

i) Cannizzaro's reaction

ii) Coupling reaction

iii) Rosemunnd's reduction

iv) Hydroboration-oxidation

v) Williamson's etherification

vi) Sandmeyer's reaction.

vii) Carbonylation reaction

viii) Perkin's reaction (8×1=8)

21. (a) Order of reaction can experimentally be determined by comparing the rate of reaction with change in concentration of particular reactant.

i) Define order of reaction.

ii) Write down the unit of rate constant of zero order reaction.

iii) Give any two points of differences between order and molecularity of reaction. (1+1+2)

b) For the following hypothetical reaction.

$M+N \rightarrow Q$

When the concentration M is doubled, rate becomes doubled. If the concentration of both M and N are doubled, rate becomes 16 times.

i) Write down rate law equations

ii) Find out the overall order of reaction

(2+2)

Or

a) During titration, the concentration of $KMnO_4$ solution can be determined by using standard oxalic acid solution.

(1+1+1+1)

i) What is meant by standard solution ?

ii) Calculate the equivalent weight of $KMnO_4$ is acidic medium.
(Molar mass of $KMnO_4$ =158)

iii) Why is above titration called redox titration ?

iv) Name the indicator used in this titration.

b) 1 gm of a divalent metal was dissolved in 25ml of $1MH_2SO_4$. The unreacted acid further required 15 C.C. of NaOH (f=0.8) for complete neutralization.
(1+3)

i) Calculate the gram equivalent of unreacted acid

ii) Find the atomic weight of metal.

22. a) What is meant by artificial radioactivity? Write an example of it.
(1+1)

b) PVC and nylon-6,6 are two common polymers widely used in daily life.

i) State the process of polymerization by which nylon-66 is formed.

ii) Write down the chemical reaction to form PVC. (1+1)

c) i) What is Portland cement? Name the major compositions of Portland cement.
(1+2)

ii) Define the term clinker
(1)